Magnetism and Matter

DISCLAIMER

"The content provided herien are created and owned by various authors and licensed to Sorting Hat Technologies Private Limited ("Company"). The Company disclaims all rights and liabilities in relation to the content. The author of the content shall be solely responsible towards, without limitation, any claims, liabilities, damages or suits which may arise with respect to the same."

Magnetic Poles and Bar Magnet

We have seen that a small current loop carrying a current i, produces a magnetic field at an axial point is-

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{2\vec{\mu}}{d^3} \qquad ...(1)$$

where $\mu = i\vec{A}$ = It is defined as magnetic dipole moment of the current loop,

$\vec{A}$ = area-vector of the current loop.

Also, a current loop experiences a torque when placed in a magnetic field $\vec{B}$ -

$$\vec{\tau} = \vec{\mu} \times \vec{B} \qquad ...(2)$$

Similarly, an electric dipole produces an electric field at an axial point is-

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}.\frac{2\vec{p}}{d^3} \qquad ...(3)$$

and it experiences a torque when placed in an electric field is

$$\vec{\tau} = \vec{p} \times \vec{E} \qquad ...(4)$$

equation (1) and (2) for a current loop are similar in structure to the equation (3) and (4) for an electric dipole with $\vec{\mu}$ taking the role of $\vec{p}$ and $\frac{\mu_0}{4\pi}$ taking the role of $\frac{1}{4\pi\varepsilon_0}$.

Hypothetical model related to magnetic charge, force, and field:

(a) Similar to electrostatics, hypothetically it can be assumed that there are two magnetic charges (North pole as positive magnetic charge and south pole as negative magnetic charge) the magnitude of each is represented by m. On keeping a magnetic charge in a magnetic field, the force acting on it is given by

$$\vec{F} = m\vec{B} \qquad ...(1)$$

The direction of force on the magnetic charge will be along the direction of magnetic field vector for positive magnetic charge (North pole) and the direction of force will be in opposite direction of the magnetic field vector for negative magnetic charge (south pole).

(b) Magnetic field produced by a point magnetic charge at a distance r is given by:

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{m}{r^2}\hat{r} \qquad ...(2)$$

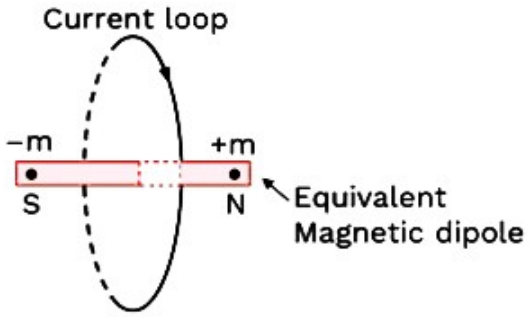
(c) On keeping a positive and negative magnetic charge very close to each other, a magnetic dipole is formed.

If d is the separation between the magnetic charges, then magnetic dipole moment is given by, M = md.

$$\vec{M} = m\vec{d} \qquad ...(3)$$

The magnetic moment $\vec{M}$ of a bar magnetic is very similar with $\vec{\mu}$ in current loop.

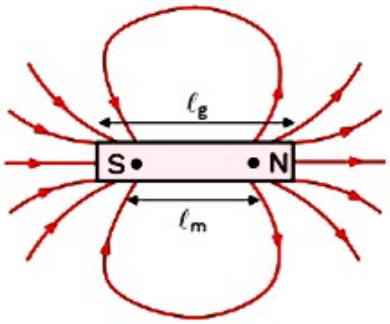
(d) Suppose a current-carrying loop has current I and area A. This current-carrying loop may be replaced by a magnetic dipole. Its dipole moment md = IA. The area vector $\vec{A}$ points in the direction −m to +m.



(e) It is customary to call a +ve magnetic charge a "The North Pole" and a −ve magnetic charge a "South Pole." The quantity m is called "pole strength." As md = iA, the unit of m will be A-m (Ampere-meter).

Bar Magnet:

A bar magnet is a rectangular or cylindrical piece of an object made up of steel, iron, or any other ferromagnetic material. The poles of a bar magnet are slightly inside the two ends. The line joining the positions of the assumed poles is called the "magnetic axis" of the bar magnet.



At the two ends of a natural magnet, the field behaves differently from those inside the magnet. Because of this effect, the magnetic poles appear slightly inside the bar. The distance between the locations of the assumed poles is called the "magnetic length" of the magnet. The distance between the ends is called the "geometrical length." It is found that-

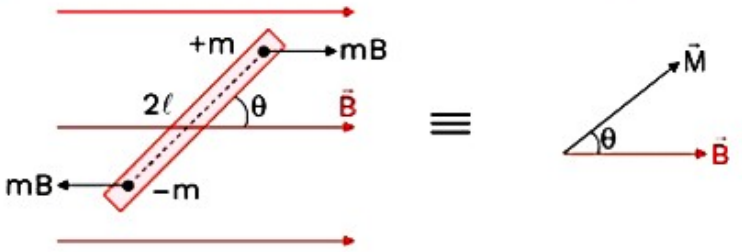
$$\frac{\text{magnetic length}}{\text{geometrical length}} \approx 0.84 \quad \Rightarrow \frac{\ell_m}{\ell_g} \approx 0.84$$

If m be the pole strength and 2ℓ be the magnetic length of a bar magnet, its magnetic moment M is-

$$M = 2m\ell$$

Torque on a Bar Magnet

Calculation of Torque on a Bar Magnet Placed in a Uniform Magnetic Field:



The torque due to both forces-

$\tau = mB(2\ell\sin\theta)$ $\Rightarrow \vec{\tau} = (2m\ell)\, B\sin\theta$ (clockwise)

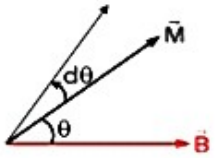
$\Rightarrow$ $\boxed{\vec{\tau} = \vec{M} \times \vec{B}}$

This torque tries to rotate the magnet so as to align it with the field.

Potential Energy of a Bar Magnet Placed in a Uniform Magnetic Field

Potential Energy of a Bar Magnet Placed in a Uniform Magnetic Field:

We can calculate Potential energy of bar magnet by calculating the work done by the external agent in changing the angle slowly from θ to $\theta + d\theta$ is-



$$dW_{ext} = \vec{\tau}_{ext}.d\vec{\theta} = -\vec{\tau}_{mag}.d\vec{\theta}$$

$$dU = MB \sin\theta \, d\theta \qquad ...(1)$$

Now the change in potential energy in rotating the magnet from an initial angle θ_i to final angle θ_f is-

$$\int_i^f dU = MB\int_{\theta_i}^{\theta_f} \sin\theta \, d\theta$$

$$\boxed{U_f - U_i = -MB(\cos\theta_f - \cos\theta_i)} \qquad ...(2)$$

If we take $\theta = 90°$ as a reference point and assign $U_{90°} = 0$, the potential energy at an angle θ is-

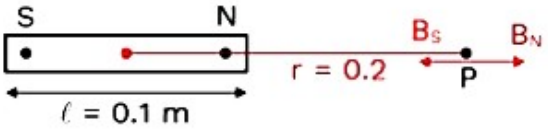
$$U_\theta - U_{90°} = -MB(\cos\theta - \cos 90°) = -MB\cos\theta$$

$$\boxed{U_\theta = -\vec{M}.\vec{B}} \qquad ...(3)$$

from equation (3), SI unit of magnetic moment M may also be written as JT^{-1}.

Q. **A bar magnet is 0.1 m long, and its pole strength is 12 A-m. Calculate the magnetic field at a point on its axis at a distance of 0.2 m from its center.**

Sol. Net magnetic induction at point P,



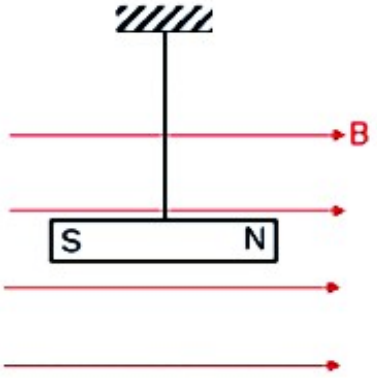
$$B_P = B_N - B_S = \frac{\mu_0}{4\pi}.\frac{m}{\left(r-\frac{\ell}{2}\right)^2} - \frac{\mu_0}{4\pi}.\frac{m}{\left(r+\frac{\ell}{2}\right)^2}$$

$$B_P = 10^{-7} \times 12\left[\frac{1}{(0.2-0.05)^2} - \frac{1}{(0.2+0.05)^2}\right]$$

$$B_P = 12 \times 10^{-7}\left[\frac{1}{0.15^2} - \frac{1}{0.25^2}\right] = 12 \times 10^{-7}[44.44 - 16]$$

$B_P = 34.13\ \mu T$ along the axis of dipole

Q. **Suppose a magnetic dipole of magnetic moment M is suspended by a string in a uniform horizontal magnetic field. If in horizontal plane this dipole is slightly tilted and released, prove that it will execute SHM and find its oscillation period. The bar magnet is a uniform rod of mass m and length l.**



Sol. If dipole is tilted slightly (angle θ) from equilibrium position, the restoring torque on it is-

$$\vec{\tau} = -MB\sin\theta\,\hat{\theta} \quad \Rightarrow \quad \tau \approx -MB\theta$$

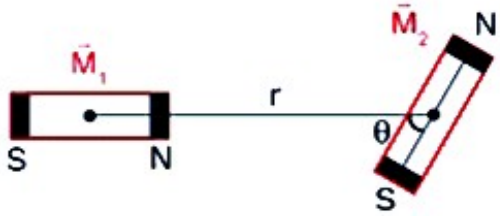
Angular acceleration α is-

$$\alpha = \frac{\tau}{I} = \frac{-MB\theta}{(m\ell^2/12)} = -\left(\frac{12MB}{m\ell^2}\right)\theta = -\omega^2\theta$$

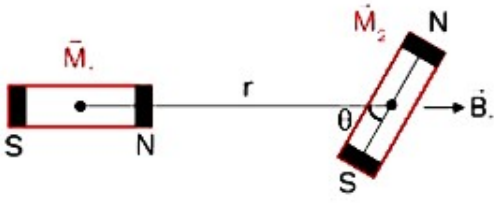
we can see that torque is linearly proportional to θ, therefore rod will execute SHM.

$$\Rightarrow \quad \omega = \sqrt{\frac{12MB}{m\ell^2}} \quad \Rightarrow \quad \boxed{T = 2\pi\sqrt{\frac{m\ell^2}{12MB}}}$$

Q. **The figure shows two small bar magnets placed at separation r. Find the magnetic interaction energy of this system of dipoles.**



Sol. Field due to $\vec{M}_1$ at the location of $\vec{M}_2$ is-



$$\vec{B}_1 = \frac{\mu_0}{4\pi}\frac{2\vec{M}_1}{r^3}$$

Interaction energy of $\vec{M}_2$ in the field of $\vec{M}_1$,

$$U = -\vec{M}_2.\vec{B}_1 = -M_2 B_1 \cos\theta$$

$$\boxed{U = \frac{-\mu_0 M_1 M_2 \cos\theta}{2\pi r^3}}$$

Q. **Suppose a bar magnet had a magnetic moment of 2.0 Am². The bar magnet is free to rotate about a vertical axis through its center. The magnet is in the East-West position and released from the rest from. Calculate the kinetic energy of the magnet as it takes the North-South position. It is given that The horizontal component of the Earth's magnetic field is B = 25 μ T.**

Sol. Energy released = $-W_{ext} = -\Delta U$

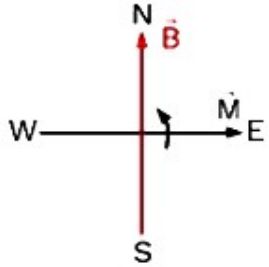
Energy released = $-MB(\cos\theta_i - \cos\theta_f)$

$E = -MB(\cos 90° - \cos 0°)$

$E = MB$

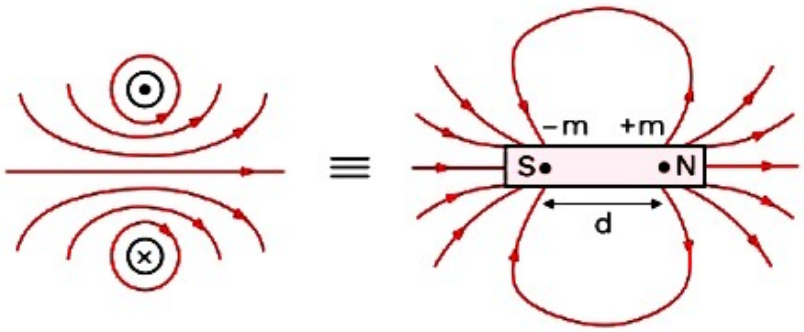
$E = 2 \times 25\ \mu$

$E = 50\ \mu J$

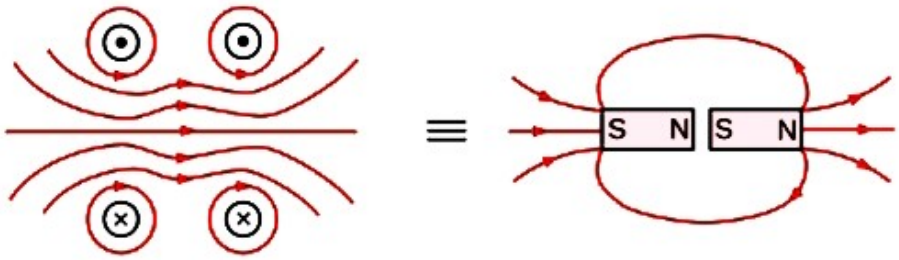


Solenoid as a Bar Magnet

A current loop is equivalent to a short magnetic dipole with negligible separation (d) between the poles.



Similarly, two nearly placed current loops is equivalent to as following-



We can treat a solenoid very closely resembles a combination of circular loops placed side by side, as shown in figure 1. If I be the current through flowing through the solenoid and A be the area of cross-section, the dipole moment of each turn is μ = iA. In this model, each turn may be replaced by a small dipole (small bar magnet) placed at the center of the loop along its axis (Figure 2).

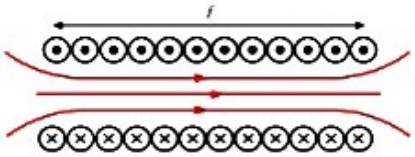


Figure 1

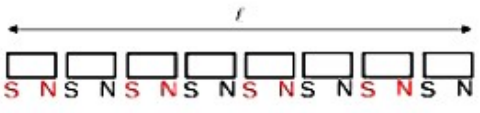


Figure 2

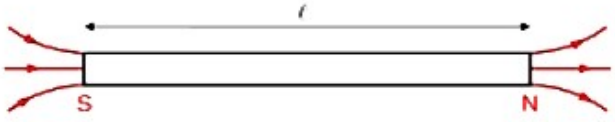


Figure 3

Suppose each turn is replaced by a magnetic dipole with pole strength m and separation d between the poles (Figure 2). We have-

$$M = \mu \quad \Rightarrow \quad md = iA$$

On replacing the circular current loops by small magnetic dipoles, we can see that the alternate south and north poles cancel out each other's effect leaving only the poles at the ends to be active.

So, it can be said that the solenoid can be replaced by a bar magnet of length equal to the length of the solenoid with a single south and single north pole having a pole strength m as shown in the figure 3

$$\mu_{solenoid} = M_{bar\ magnet}$$

$$NiA = m\ell$$

$$m = \frac{NiA}{\ell}$$

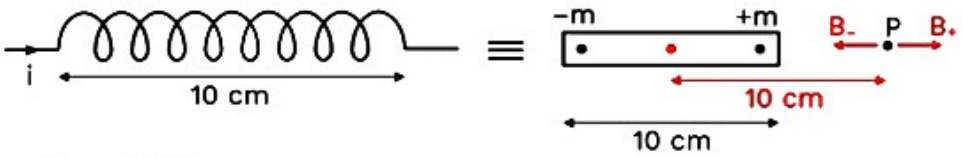
where N = number of turns in solenoid;

ℓ = length of solenoid = length of bar magnet;

m = magnetic pole strength.

Q. **A solenoid of radius 1 cm and length 10 cm contains 200 turns and carries a current of 10 A. Find the magnetic field at a point on the axis at a distance of 10 cm from the center.**

Sol.



$\because$ $\mu_{solenoid} = M_{equivalent\ bar\ magnet}$

$$NiA = m\ell$$

$$m = \frac{200 \times 10 \times \pi(10^{-2})^2}{0.1}$$

$$m = 2\pi \text{ A-m}$$

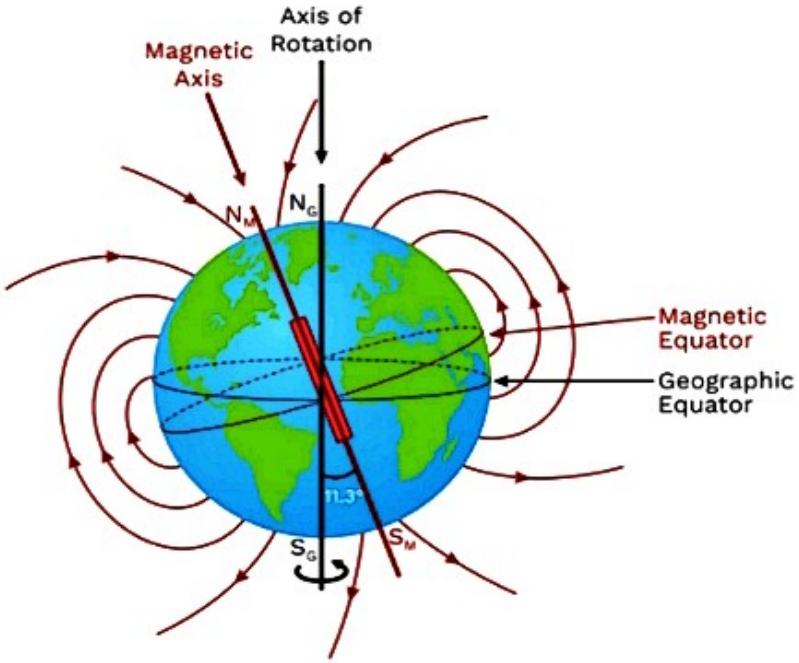
The field at P,

$$B_p = (B_+) - (B_-) = \frac{\mu_0}{4\pi} m \left[\frac{1}{(0.1-0.05)^2} - \frac{1}{(0.1+0.05)^2}\right]$$

$$B_p = 10^{-7} \times 2\pi\,[400 - 44.44]$$

$$B_p = 2.23 \times 10^{-4}\ T \text{ (away from the solenoid)}$$

The Earth's Magnetism



The magnitude as well as direction of Earth's magnetic field changes with position on Earth's surface as well as time.
The variation of Earth's magnetic field is not noticeable over a span of few days but in a time span of centuries, there can be noticeable changes.
The magnetic field of earth varies from place to place on its surface in magnitude as well as in direction. The magnitude of Earth's magnetic field is of the order of 10^{-5} T.
Studies of magnetic rocks by geologists suggest that the magnetic field of Earth can also change its direction. Such phenomenon is called field reversal. Data from studies suggests that in past 7.6×10^7 years, 171 field reversals have already occurred. The last field reversal was around 10000 years ago.

Dynamo Effect

For a long time, it was hypothetically assumed that the magnetic field of Earth is due to a giant bar magnet kept at the centre of Earth with its axis aligned to the axis of Earth's rotation. However, this assumption is for simplification and it is certainly not correct.
It is now considered that the magnetic field of Earth is arising due to electric currents generated inside earth due to convective motion of iron and nickel (in molten form) in the outer core of Earth. This is termed as dynamo effect.
The magnetic dipole-axis of Earth makes an angle of about 11.3° with the Earth's axis of rotation.
The pole near the geographic N-pole (N_G) is called the "Geo-magnetic N-pole (N_M)." Similarly, the pole near the geographic S-pole (S_G) is called "geomagnetic S-pole (S_M)".
The nomenclature of the poles has been done on the basis that the magnetic north was the direction to which the north pole of a magnetic pole behaves like the south pole of a bar magnet inside the Earth and vice-versa.

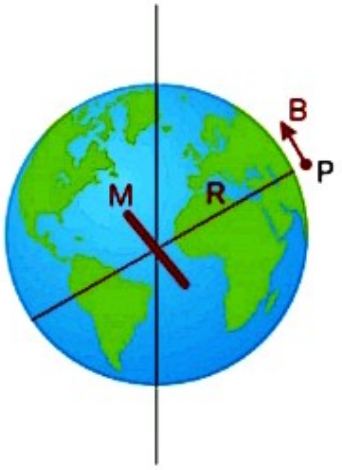
Q. **Find the dipole moment of Earth if the magnetic field at the equator of Earth is 0.4 Gauss.**

Sol. It can be assumed that the magnetic field of Earth is due to a small bar magnet of magnetic moment M kept at the centre of Earth.
Point P on Earth's surface is on the magnetic equator and far away from the magnet.

$$B_P = B_{equitorial} = \frac{\mu_0}{4\pi}\frac{M}{R^3}$$

$$0.4 \times 10^{-4} = 10^{-7}\frac{M}{(6.4 \times 10^6)^3}$$

$$\boxed{M = 1.05 \times 10^{23}\ A - m^2}$$

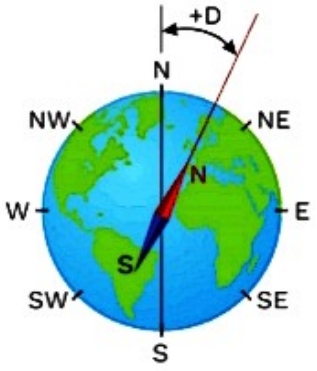


Elements of the Earth's Magnetic Field

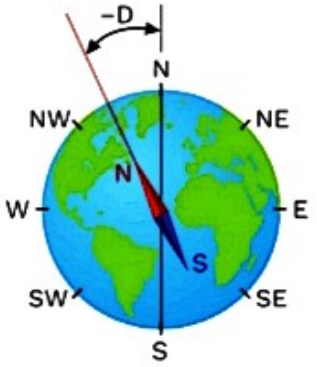
At any point on the surface of Earth, the magnetic field can be completely described by 3 quantities:

(a) Magnetic Declination (D)
(b) Magnetic Inclination or Dip angle (δ)
(c) Horizontal component of net field (B_h)

(a) Declination (D):

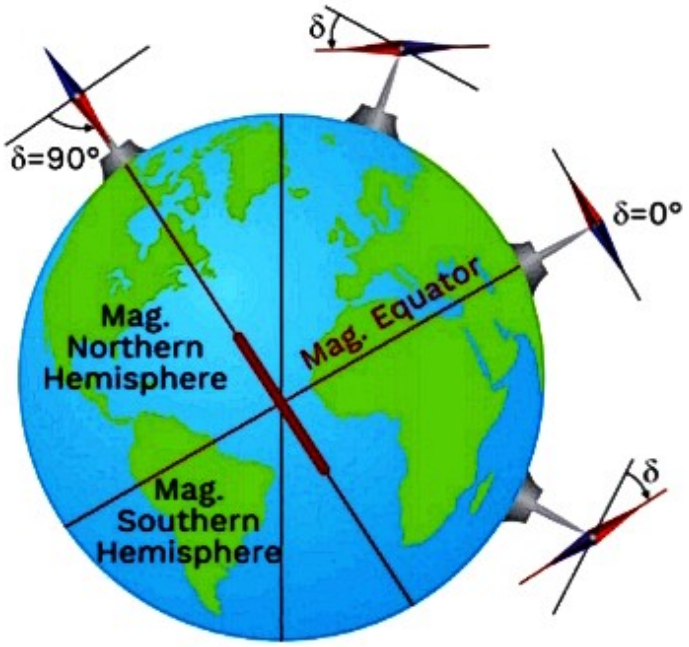


A plane which passes through the geographical poles and also through a given point P on the surface of Earth is known as "geographical meridian" at that point P.
Similarly, the plane which passes through the geomagnetic poles (i.e., through the dipole axis of Earth) and the point P is known as the "magnetic meridian" at the point P.
The angle between the magnetic meridian at a point with the geographical meridian is called the "declination" at that point.
If the compass at your place is pointing "clockwise" w.r.t. the true north, declination is +ve or EAST.



If the compass at your place is pointing counter-clockwise w.r.t the true-north, declination is –ve or WEST.
The magnetic declination in India is small, it being 0°41' E at Delhi and 0°58' W at Mumbai.

(b) Inclination or dip (δ):



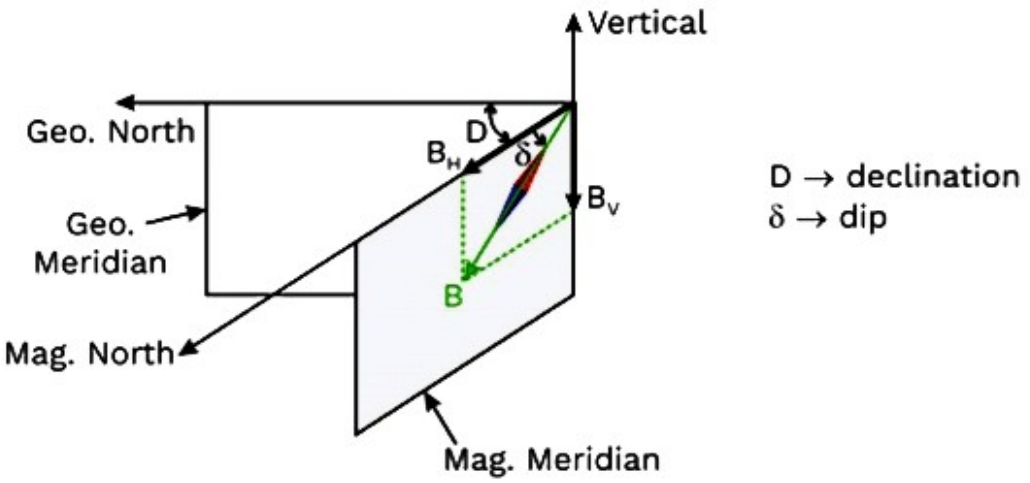
The angle between the net magnetic field vector of Earth and the horizontal component of Earth's magnetic field in the magnetic meridian at any point on Earth's surface is called "magnetic inclination" or "dip" at that point.

In the magnetic northern hemisphere, the vertical component of the Earth's magnetic field points downwards. The north pole of a freely suspended magnet, therefore, dips (goes down). Similarly, in the magnetic southern hemisphere, the vertical component of the Earth's magnetic field points upwards. The north pole of a freely suspended magnet, therefore, goes up (inclination).

The knowledge of declination and inclination completely specifies the direction of the Earth's magnetic field.

(c) Horizontal Component of the Earth's Magnetic Field:

As the name indicates, the "horizontal component" is the component of the Earth's magnetic field in the horizontal direction in the magnetic meridian. This direction is towards the magnetic north.



The above figure is indicating the three elements of Earth's magnetic field at a point-

(i) Magnetic meridian is drawn making an angle D with the geographical meridian.

(ii) The horizontal direction which describes the magnetic north is drawn in the magnetic meridian. The net magnetic field is making an angle δ with the horizontal direction.

(iii) The horizontal component of magnetic field B_H is drawn as shown in figure. The relation between net magnetic field (B) and the horizontal component (B_H) is given as-

$$B_H = B\cos\delta$$

$$B = \frac{B_H}{\cos\delta}$$

Q. **Given that the horizontal component of the Earth's magnetic field is 3.6×10^{-5} T, where the dip angle is 60°. Find the magnitude of Earth's magnetic field.**

Sol. $B_H = B\cos\delta \Rightarrow B = \frac{B_H}{\cos\delta} = \frac{3.6\times10^{-5}}{\cos 60°}$

$$\boxed{B = 7.2\times10^{-5}\ T}$$

Q. **A short bar magnet is placed in a magnetic meridian with its N-pole pointing towards the south. Its magnetic moment is 0.2 A-m². The neutral point is obtained 10 cm from the center of the magnet toward the north. Calculate the horizontal component of Earth's magnetic field.**

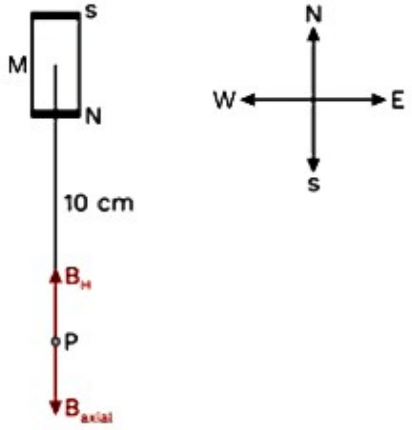
Sol. ∵ $B_H = B_{axial}$ (at point P)

$$B_H = \frac{\mu_0}{4\pi}\frac{2M}{r^3}$$

$$B_H = 10^{-7} \times \frac{2 \times 0.2}{(0.1)^3}$$

$$B_H = 4 \times 10^{-5}\ T$$

$$\boxed{B_H = 40\ \mu T}$$



Q. **A short magnet of magnetic moment 6 A-m² is lying in a horizontal plane with its north pole pointing toward 60° East of North. Find the magnitude of the net horizontal magnetic field at a point on the axis of the magnet 0.2 m away from it. The horizontal component of the earth field at this place is 30 μ T.**

Sol. The whole system lie on horizontal plane

$B_H = 30\ \mu T$

$$B_{axial} = \frac{\mu_0}{4\pi}\frac{2M}{r^3}$$

$$B_{axial} = 10^{-7} \times \frac{2 \times 6}{(0.2)^3}$$

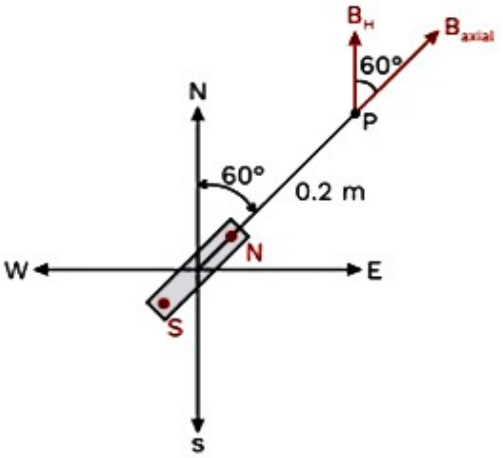
$$B_{axial} = \frac{1200}{8}\mu T$$

$$B_{axial} = 150\ \mu T$$

$$\therefore \quad B_P = \sqrt{B_{axial}^2 + B_H^2 + 2B_{axial} \cdot B_H \cos 60°}$$

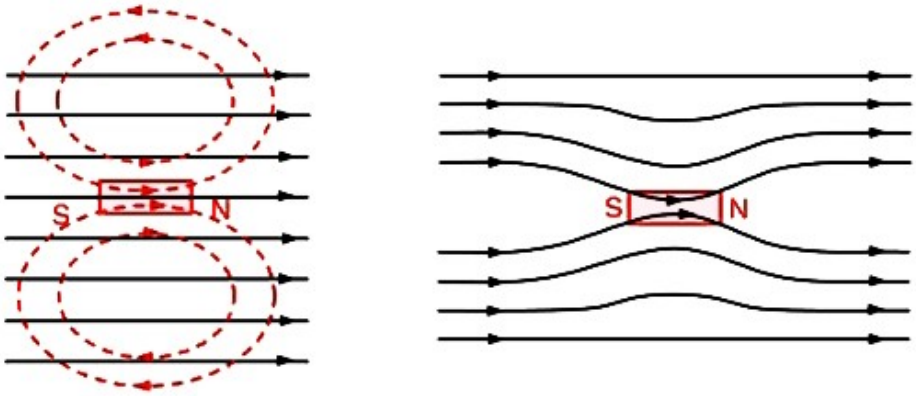
$$B_P = \sqrt{150^2 + 30^2 + 2(150)(30)\left(\frac{1}{2}\right)}\ \mu T$$

$$\boxed{B_P = 167\ \mu T}$$



Some Important Terms Used in Magnetism

(A) Magnetic Induction ($\vec{B}$):



On placing a piece of substance in an external magnetic field, the substance gets magnetised.

When an iron bar is placed in a uniform magnetic field, the bar gets magnetised and it produces its own magnetic field.

The direction of the magnetic field produced by the magnetised bar has same direction as that of the applied uniform magnetic field inside the bar while its direction is opposite to the external field outside the bar.

Due to these directions of the magnetic field due to the magnetized bar, the magnetic lines of force get concentrated within the bar. The magnetic flux density increases inside the bar and decreases at some places outside the bar.

"The number of magnetic lines of induction inside a magnetic substance crossing unit area normal to their direction is called the magnitude of magnetic induction, or magnetic flux density inside the substance." It is denoted by $\vec{B}$.

(B) Intensity of Magnetisation ($\vec{I}$):

"Intensity of magnetisation ($\vec{I}$) is defined as the magnetic moment per unit volume of the magnetised substance." Intensity of magnetisation represents the extent to which a material is magnetised on placing in an external magnetic field. Thus,

$$\vec{I} = \frac{\vec{M}}{V} \qquad \ldots(1)$$

where $\vec{M}$ = magnetic moment

V = volume of substance

The SI unit of $\vec{I}$ is ampere/meter (A/m).

(C) Magnetic Intensity or Magnetic Field Strength ($\vec{H}$):

On placing a substance in an external magnetic field, the net magnetic field inside the substance is sum of the external magnetic field and the magnetic field due to the magnetisation of the substance.

Now, a quantity "magnetic intensity" or "magnetizing field intensity" ($\vec{H}$) is defined as-

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{I} \qquad ...(2)$$

where $\vec{B}$ = resultant magnetic field inside the material,

$\vec{I}$ = intensity of magnetization of that material.

The unit of H is same as that of I, that is A/m.

If no material is present (vacuum), $\vec{I} = 0$ & we have,

$$\vec{H} = \frac{\vec{B}}{\mu_0} \qquad ...(3)$$

Q. **A bar magnet has a mass of 6.6×10^{-3} kg and magnetic moment of 2.5 Am^2. If the density of the bar magnet is 7.9×10^3 kg/m^3, find the intensity of magnetization of the magnet.**

Sol.

$$V = \frac{\text{mass}}{\text{volume}} = \frac{6.6 \times 10^{-3}}{7.9 \times 10^3} = 8.3 \times 10^{-7}\ m^3$$

So, $$\left|\vec{I}\right| = \frac{m}{V} = \frac{2.5\ Am^2}{8.3 \times 10^{-7}\ m^3} = 3 \times 10^6\ A/m.$$

Q. **For a long solenoid carrying current i and having n turns per unit length, find the magnetic intensity H at its centre when,**
(a) when there is no material inside the solenoid, and
(b) when a long copper rod is inserted in the solenoid.

Sol. (a) When no material is kept inside the solenoid, magnetic field at the centre of solenoid is-

$$B = \mu_0 ni$$

The magnetic intensity is-

$$H = \frac{B}{\mu_0} = ni$$

(b) Placing a rod inside the solenoid has no effect on the magnetic intensity of the center. It has same value in both the cases. So, H = ni.

(D) Magnetic Susceptibility:

For paramagnetic & diamagnetic materials, the intensity of magnetisation of a material is directly proportion to the magnetic intensity. Thus,

$$\vec{I} = \chi \vec{H} \quad \ldots(6)$$

The proportionality constant χ is called the susceptibility of the material. As I & H have the same dimensions, the susceptibility χ is a dimensionless constant.
As there can be no magnetisation in vacuum, I = 0 & hence $\chi = 0$. The materials with +ve value of χ are paramagnetic & those with negative value of χ are diamagnetic.

Susceptibilities of some materials:

Material	Temp. in °C	χ (in 10^{-5})
Vacuum	-	Zero
Air	STP	0.04
O_2	STP	0.18
Mg	20	1.2
Al	20	2.1
Tungsten	20	6.8
Titanium	20	7.06
Cerium	18	130
Ferric Chloride	20	306
Oxygen (liquid)	−219	490
CO_2 (1 atm)	20	-2.3×10^{-4}

Material	Temp. in °C	χ (in 10^{-5})
Nitrogen (1 atm)	20	-5.0×10^{-4}
Hydrogen (1 atm)	20	-9.9×10^{-4}
Sodium	20	−0.24
Copper	18	−0.96
Lead	18	−1.6
Carbon (diamond)	20	−2.2
Carbon (graphite)	20	−9.9
Gold	20	−3.6

(E) Permeability:

The magnetic intensity is given by-

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{I}$$

or $$\vec{B} = \mu_0(\vec{H} + \vec{I})$$

$$\vec{B} = \mu_0(\vec{H} + \chi\vec{H})$$

$$\vec{B} = \mu_0(1 + \chi)\vec{H}$$

or we can write $\vec{B} = \mu\vec{H}$ ...(7)

where $\mu = \mu_0(1 + \chi)$ is a constant called the "permeability" of the material. (The permeability of vacuum is μ_0 as $\chi = 0$ for vacuum).

The constant $\mu_r = \frac{\mu}{\mu_0} = 1 + \chi$ ...(8)

μ_r is called the relative permeability of the material.
We can understand the significance of relative permeability by the following example. Let us consider a long solenoid (or a toroid) carrying a current i and having n turns per unit length. The magnetic field B0 inside the solenoid is-

$$B_0 = \mu_0 ni$$

& the magnetic intensity is $H = ni$.
Now, suppose, a material is inserted into the solenoid. The magnetic field now becomes-

$$B = \mu H = \mu ni = \mu_r \mu_0 ni = \mu_0(1 + \chi) ni$$

& $$\frac{B}{B_0} = \frac{\mu}{\mu_0} = \mu_r$$

Thus, μ_r is the factor by which the magnetic field B is increased when a material is brought in the field.]

Q. **Find the % increase in the magnetic field B when the space within a current carrying toroid is filled with aluminium. The susceptibility of aluminium is 2.1 × 10^{-5}.**

Sol. In absence of Al, the magnetic field is-

$$B_0 = \mu_0 H$$

As the space inside the toroid is filled with Al, the field becomes-

$$B = \mu H = \mu_0(1 + \chi)H$$

The increment in the field is $B - B_0 = \mu_0 \chi H$ % increament is-

$$\frac{B - B_0}{B_0} \times 100 = \frac{\mu_0 \chi H}{\mu_0 H} \times 100 = (\chi) \times 100 = 2.1 \times 10^{-3}\ \%$$

Properties of Magnetic Materials

Based on the magnetic properties of substances, we can classify all materials (solids, liquids or gases) into three categories:

(i) Paramagnetic materials
(ii) Diamagnetic materials
(iii) Ferromagnetic materials

(i) Paramagnetic materials:

Magnetic properties of paramagnetic materials are:

(a) The materials which acquire a small or feeble magnetisation when placed in an external magnetic field in the same direction as that of the external magnetic are called paramagnetic materials.

As the external magnetic field and the magnetic field due to the magnetisation are in same direction, so the magnetic induction inside the substance is slightly greater than outside to it. (Figure 1)

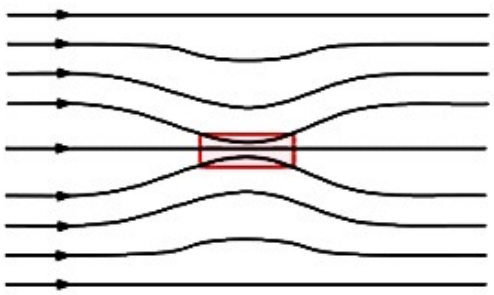


Figure 1 (For paramagnetic material)

(b) These materials rotate in a uniform magnetic field until their longest axis is parallel to the field.

(c) On placing in a non-uniform magnetic field, these materials are attracted towards regions of the stronger magnetic field.

(d) $0 < \chi < \varepsilon$

$1 < \mu_r < 1 + \varepsilon$

$\mu > \mu_0$

where ε is a small positive number.

Few paramagnetic materials are Platinum, Aluminium, Chromium, $CuSO_4$ solution, etc.

(ii) Diamagnetic materials:

(a) The materials which acquire a small or feeble magnetisation when placed in an external magnetic field in the opposite direction as that of the external magnetic are called diamagnetic materials.

As the external field and the field due to magnetisation are in opposite directions, so the magnetic field inside the material is smaller than the external magnetic field.

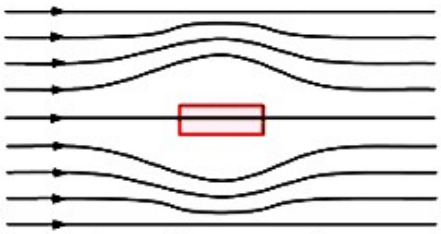


Figure 2 (For diamagnetic material)

(b) In a uniform field, these materials rotate until their longest axis are normal to the field.

(c) These materials move from stronger to weaker part of the magnetic field when placed in a non-uniform magnetic field.

(d) $-1 \leq \chi < 0$

$0 \leq \mu_r < 1$

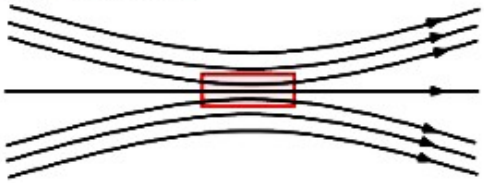
$\mu < \mu_0$

(e) Some diamagnetic materials are Type 1 superconductors as they exhibit perfect conductivity and perfect diamagnetization ($\chi = -1$, $\mu_r = 0$) when cooled to very low temperature. A type 1 superconductor repels a magnet, and by Newton's 3rd Law, it is repelled by the magnet. The phenomenon of perfect diamagnetism in superconductors is called the "Meissner Effect."

Few diamagnetic materials are Bismuth, Gold, Quartz, Antimony, Water, Alcohol, etc.

(iii) Ferromagnetic materials:

(a) The materials which get magnetised strongly even by a weak external magnetic field are called ferromagnetic materials.



(b) $\chi >> 1$

$\mu_r >> 1$

$\mu >> \mu_0$

(c) With increase in temperature, magnetic susceptibility steadily decreases. Ferromagnetic materials becomes paramagnetic above a certain temperature known as "Curie temperature" (T_c).

Value of Curie's temperature is 631 K for nickel and 1043 K for iron.

when $T < T_c$	when $T > T_c$
$\chi = \frac{C}{T}$, C is constant.	$\chi = \frac{C'}{T - T_c}$, C' is a constant.

Few ferromagnetic materials are Iron, Nickel, Steel, Cobalt, etc.

Explanation of Paramagnetism, Diamagnetism and Ferromagnetism

The magnetic dipole moment in materials is due to three properties of atoms of the materials:

(1) The revolution of electrons around nucleus in orbits can be seen as small current loops which produce magnetic moment known as orbital magnetic moment (μ_o).

(2) An intrinsic magnetic dipole moment (μ_i) of spinning electrons.

(3) The motion of charges withing the nucleus also produces magnetic moment.

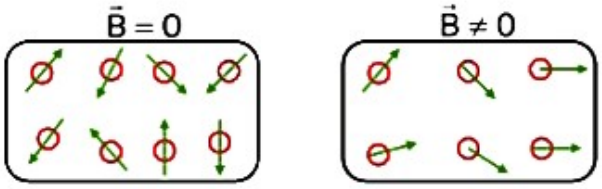
A large value of the magnetic moment of an atom is due to the intrinsic magnetic dipole moment of the spinning electrons. We know from molecular orbital theory that the electrons form pairs with electrons of opposite spin. Due to the opposite spin of pairing electrons, the orbital dipole moments cancel out.

The intrinsic moment is not always possible to cancel out as formation of equal and opposite pairs is not always possible.

(i) Paramagnetism:

Paramagnetism is seen in the materials with atoms or molecules having excess electrons spinning in the same direction. Paramagnetic atoms act as tiny bar magnets and have permanent magnetic moments.

When external magnetic field is absent, the atomic magnets are randomly oriented such that the net magnetic moment is zero.



Tiny Bar Magnets

On applying an external magnetic field on a paramagnetic material, each tiny atomic magnet rotates in the direction of the applied magnetic field and gets aligned in its direction. In this way, the whole material is magnetised in the same direction of the external magnetic field.

Paramagnetism property of material depends on its temperature as the increase in temperature changes the magnetic alignment of atoms.

(ii) Diamagnetism:

Diamagnetism is seen in the materials having paired electrons in the orbitals. Total magnetic moment of diamagnetic materials is zero.

On placing the diamagnetic materials in an external magnetic field, the spin of the electrons changes. The electrons which produce magnetic moment in opposite direction of external magnetic field are accelerated while electrons with the magnetic moment in the direction of the external magnetic field are decelerated.

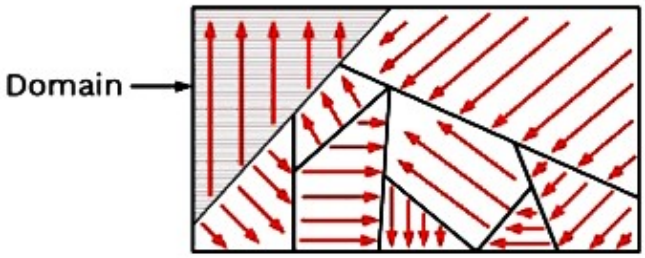
The net magnetic moment is opposite to the direction of external magnetic field and the material has a magnetisation in opposite direction of external field.

Property of diamagnetism does not depend upon the temperature of the material.

(iii) Ferromagnetism:

Ferromagnetic materials have a very high (of the order of 10^3) magnetic susceptibility. Atoms of ferromagnetic materials have permanent magnetic moments and the atoms form a large number of small effective regions called "domains".

The domain size ranges from 10^{-6} cm^3 to 10^{-2} cm^3 and each domain contains 10^{17} to 10^{21} atoms having the magnetic moments aligned in same direction.

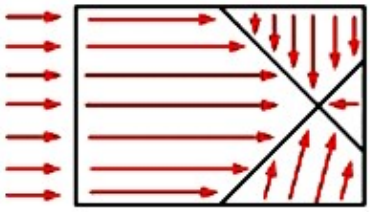


Unmagnetised

In absence of an external magnetic field, the tiny magnets have random orientation making the net magnetisation to be zero.

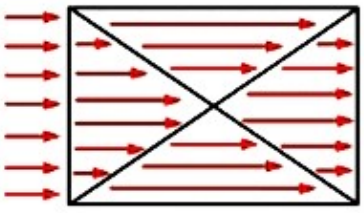
On placing the ferromagnetic materials in external magnetic field, the magnetisation can occur in two ways.

(a) For a weak external magnetic field:



On placing ferromagnetic materials in weak magnetic field, the domains which produce magnetic moment in same direction of the external field increase in size while the domains having opposite magnetic moments to the external field reduce in size.

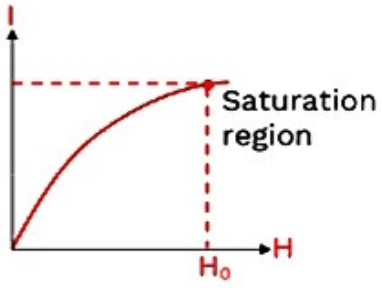
(b) For a strong external magnetic field:



On placing the ferromagnetic material in strong magnetic field, all the domains get aligned along the field direction.

Curie Law in Magnetism

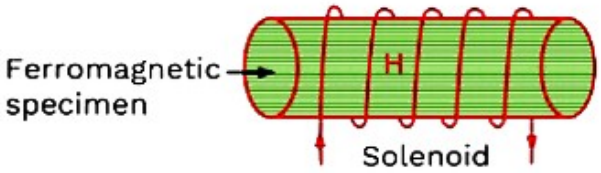
Pierre Curie experimentally discovered that intensity of magnetisation I is directly proportional to magnetic intensity H and inversely proportional to the absolute temperature (T).



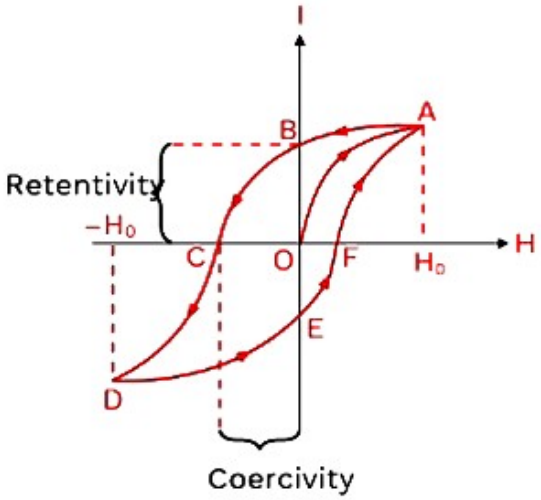
i.e., $I \propto \frac{H}{T}$ $\Rightarrow \frac{I}{H} \propto \frac{1}{T}$ $\Rightarrow \boxed{\chi = \frac{C}{T}}$

The saturation region explains that at a particular stage, all atomic dipoles present in the specimen align in the direction of the field.

Hysteresis (Shown by Ferromagnetic Substances Only)



Consider a ferromagnetic specimen being magnetized slowly by a changing magnetizing field intensity (H) which is produced by the solenoid loops.



It is found experimentally that when a ferromagnetic material is being magnetised, the magnetisation intensity (I) increases with magnetising field intensity (H) as per the graphical path O-A. But when the magnetising field intensity (H) is reduced i.e. the material is demagnetised, then the I-H graph does not follow that path A-O. It follows the path A-B i.e. on making the value of H equal to zero, I has some non zero value equal to OB.

It shows that the material is left with some magnetism. This value of I, which is non-zero when H = 0 i.e. OB, is termed as "residual magnetism" or "retentivity".

On applying the magnetising field in opposite direction, value of I decreases along the path BC till it becomes zero at point C. The value of magnetising field (OC) in reversed direction which reduces the residual magnetism to zero is called "coercivity" of the material.

If the value of H is further increased in reverse direction, the I-H curve acquires a saturation at point D along the path CD. Now, when the direction of magnetising field is changed, the I-H curve follows the path DEFA.

The closed loop A-B-C-D-E-F-A thus formed is termed as the hysteresis loop. Hysteris loop describes the cycle of magnetisation of a specimen of the material and it shows how the intensity of magnetisation (I) lags behind the magnetising field (H).

Hysteresis:

The phenomenon of mismatch in the variation of magnetisation intensity while increasing and decreasing the magnetising field H is called "hysteresis" and the loop formed in the figure is called "hysteresis loop".

Conclusions:

(i) After removing the magnetisinig field, the remaining magnetisation in a material is the "Retentivity" of the material.

(ii) The reverse magnetising field needed to reduce the residual magnetism to zero is the "Coercivity" of the material.

(iii) The energy loss (thermal energy produced) per unit volume of a substance in a complete cycle of magnetization is proportional to the area of the hysteresis loop.

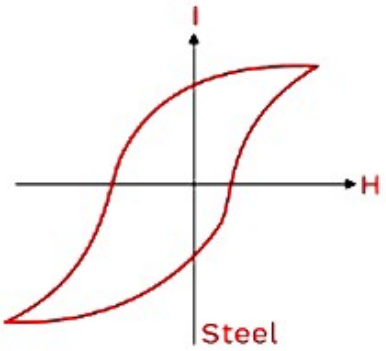
Note: The shape and size of the hysteresis loop (i.e., the value of retentivity and coercivity) are characteristic of each material, or in other words, we can say that the choice of magnetic material for different uses is decided from the hysteresis curve of a specimen of the material.

Two Types of Magnetic Material

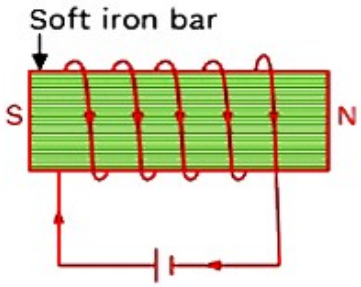
(i) **Permanent Magnets :**

Permanent magnetic materials have a large area of the hysteresis loop. To magnetise these materials, large magnetising fields are required. These materials have high retentivity and high coercivity and thus the magnetic properties of these materials do not change on removing the magnetising field or reversing the magnetising field.

Some permanent magnetic materials are steel, cobalt and similar materials.



(ii) **Electromagnets (Temporary Magnets):** Electromagnets show magnetic properties on the application of external magnetic fields but their properties are lost as soon as the magnetising field is removed. These materials have low retentivity and low coercivity.

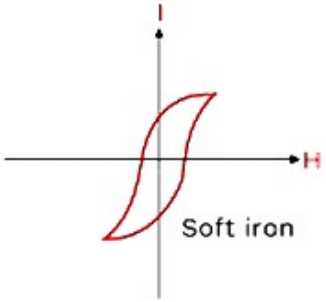


On placing a soft iron rod in a solenoid, the magnetism increase considerably and the solenoid becomes an electromagnet when a current passes through it. But as soon as the current is switched off, the magnetic properties are lost to a great extent.

For periodic variations of the magnetising fields, the energy loss in the material is very small.

Low retentivity, low coercivity and small energy loss suggest that the area of the hysteresis loop is small for electromagnets.

Soft iron is used in electric bells, cranes (to lift cars), transformers, moving coil galvanometers, etc.



Q. **A domain in ferromagnetic iron is in the form of a cube of side length 1 μm. Find the iron atoms contained in the domain and the maximum possible dipole moment and magnetization of the domain. The molecular mass of iron is 55 gm/mole, and its density is 7.9 g/cm³. Assume that each iron atom as a dipole moment of 9.27×10^{-24} Am².**

Sol. $V = 10^{-12}$ cm³, $N = 8.65 \times 10^{10}$ atoms

M is maximum when each dipole is aligned.

$M_{max} = (8.65 \times 10^{10}) \times (9.27 \times 10^{-24})$

$= 8 \times 10^{-13}$ Am²

Gauss's Law of Magnetism

It states that the total magnetic flux through any closed surface is zero. Mathematically,

$$\phi_B = \oint \vec{B}.d\vec{S} = 0$$

This theorem indicates that isolated magnetic poles do not exist or the magnetic poles always exist in pairs.